

```

struct point
{
    int x;
    int y;
}

```

```

};

```

```

struct point * p;

```

```

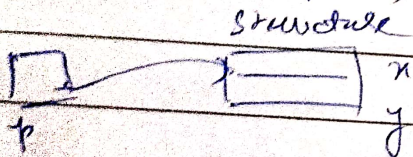
struct point p1;

```

```

p = (struct point*) malloc (size of (struct point));

```



$p \rightarrow x = 10$

$p \rightarrow y = 20$

```

struct node
{

```

```

    int val;

```

```

    struct node * next;

```

```

};

```

```

struct node * start;

```

```

start = NULL;

```

or

```

void Insertatthebeginning
(struct node **s,

```

```

    int x)

```

```

{
    struct node * p, *s;
    p = (struct node *) malloc
        (size of (struct
            node));

```

```

    p->value = x;

```

```

    p->next = *s;

```

```

    *s = p;

```

```

}

```



```

void InsertAtEnd (struct node **s, int x)
{
    struct node *p, *q;
    p = (struct node *) malloc (sizeof (struct node));
    p->value = x;
    p->next = NULL;
    if (*s == NULL)
        *s = p;

    q = *s;
    while (q->next != NULL) {
        q = q->next;
    }
    q->next = p;
}

```

```

void insert_after (struct node **s, int x, int y)
{
    struct node *p, *q;
    p = (struct node *) malloc (sizeof (struct node));
    p->value = x;
    if (*s == NULL)
    {
        p->next = NULL;
        *s = p;
    }
    else
    {
        q = *s;
        while (q->next != NULL && q->val != y)
        {
            q = q->next;
        }
        p->next = q->next;
        q->next = p;
    }
}

```

they would work in both the cases (1) insert in the middle
 (2) insert at end

Node deletion

```
void deletenode (struct node **s, int x)
{
    struct node *p, *q;
    if (*s == NULL) return;
    p = *s;
    q = NULL;
    while (p != NULL && p->value != x)
    {
        q = p;
        p = p->next;
    }
    if (q == NULL)
    {
        *s = p->next;
    }
    else if (p == NULL) return;
    else
    {
        q->next = p->next;
    }
    delete p;
}
```

Maintaining linked list in sorted order

```
void insert (struct node **s, int x)
{
    struct node *p, *q;
    p = (struct node *) malloc (sizeof (struct node));
    p->value = x;
    if (*s == NULL || x < *s->val)
    {
        p->next = *s;
        *s = p;
    }
    else
    {
        while (q->next != NULL && x < q->next->val)
        {
            q = q->next;
        }
    }
}
```



$p \rightarrow next \rightarrow q \rightarrow next;$

$q \rightarrow next \rightarrow p;$

}